

Final Inspection Procedure for Titanium Components

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Process Owner: Senior Quality Control Engineer

Purpose and Scope

This procedure establishes the standardized approach for conducting final inspections of machined titanium components to ensure compliance with aerospace and medical device industry standards, specifically AS9100 and ISO 13485, before shipment to customers.

Required Documentation and Inputs

Before beginning any final inspection, several critical inputs must be verified and assembled into a complete documentation package. The primary inputs include completed titanium parts ready for final inspection, along with the latest revision engineering drawings containing all dimensional tolerances and specifications. Each work order must detail the quantity, material grade, and specific customer requirements, accompanied by route cards showing completed manufacturing operations and in-process inspection results. Material certification including mill test certificates for titanium grade verification must be current and traceable.

Supporting documentation plays an equally critical role in ensuring thorough inspection coverage. Customer specification sheets document any additional requirements beyond drawing specifications, while previous inspection reports provide reference information when rework has been performed. Calibration certificates for all measuring equipment to be used must be current and valid, and environmental conditions logs document temperature and humidity readings throughout the inspection process.

The Final Inspection Process

Pre-Inspection Setup and Verification

The pre-inspection setup phase requires 15-20 minutes and must be performed by a Quality Technician Level II or above. This critical phase begins with inspector certification verification to confirm current qualification for titanium inspection activities. The complete documentation package review checks for completeness of all required inputs, ensuring no critical information is missing before inspection begins.

Environmental conditions must be verified to ensure temperature remains at $20^{\circ}\text{C} \pm 2^{\circ}\text{C}$ with humidity levels below 60% throughout the inspection process. Equipment calibration verification confirms all measuring tools and instruments remain within their calibration periods and are suitable for the required measurements. Component staging involves placing

parts in the designated inspection area with proper handling procedures to prevent damage or contamination.

Decision Gate A requires verification that all prerequisites are satisfied before proceeding. If any deficiencies exist, they must be documented, supervisors notified, and the process stopped until all requirements are met. Only when all prerequisites are confirmed can the inspection advance to the visual and surface inspection phase.

Visual and Surface Inspection

Visual and surface inspection requires 10-15 minutes per component and must be performed by a Certified Quality Inspector. The initial visual assessment uses 10x magnification minimum to check for machining marks, scratches, or surface defects that could affect component performance or customer acceptance. Surface finish verification ensures Ra values meet drawing requirements, while inspection for discoloration identifies potential heat damage or contamination issues.

Surface integrity verification involves blue dye penetrant testing for critical aerospace components to detect surface breaking defects invisible to visual inspection. Magnetic particle inspection may be performed if specifically required, though this is rare for titanium applications. All findings require documentation through digital photography to provide permanent records and support traceability requirements.

Decision Gate B evaluates whether surface conditions meet all specified requirements. Components meeting requirements continue to dimensional inspection, while those with surface issues require defect severity classification and determination of rework feasibility before proceeding or rejection.

Dimensional Inspection and Verification

Dimensional inspection requires 30-45 minutes per component and must be performed by a Quality Technician Level II with dimensional certification. Critical dimension verification requires Cpk values of 1.33 or higher, with all dimensions marked as "critical" on drawings measured using appropriate gauging methods. Coordinate Measuring Machines handle complex geometry while handheld instruments verify simple features, with actual measurements recorded rather than simple pass/fail determinations.

Geometric dimensioning and tolerancing verification addresses flatness, perpendicularity, concentricity, and other geometric requirements as specified on engineering drawings. Coordinate measuring machines handle complex GD&T requirements, generating comprehensive inspection reports with actual values compared to nominal specifications.

Thread inspection, when applicable, uses go/no-go thread gauges for standard threads or CMM measurement for precision thread forms, verifying thread pitch along with major and minor diameters.

Decision Gate C determines whether all dimensions meet drawing requirements. Components meeting dimensional specifications proceed to material verification, while those with dimensional issues require evaluation for rework possibility and non-conformance report generation when rejection is necessary.

Material Verification and Testing

Material verification and testing requires 20-30 minutes and must be performed by Materials Engineers or certified Quality Inspectors. Grade verification confirms titanium grades match specifications using XRF analyzer technology, while heat treatment condition verification ensures annealed, solution treated, or aged conditions meet requirements. Material certifications are checked against actual components to ensure proper traceability and compliance.

Hardness testing, when specified, uses Rockwell C scale for Grade 5 titanium typically targeting 32-38 HRC, or Brinell hardness measurements for softer grades. Minimum three readings are required with average values reported for statistical analysis. Density verification for critical applications uses hydrostatic weighing methods for complex shapes, comparing results to theoretical density values such as Grade 5 at 4.43 g/cm³.

Decision Gate D evaluates whether materials meet all specified requirements. Components meeting material requirements proceed to functional testing, while those with material issues require quarantine and investigation of material sources to prevent recurrence.

Functional Testing and Performance Verification

Functional testing timing varies based on component complexity and requirements, performed by Test Technicians under Quality Engineer supervision. Fit and assembly verification tests components with mating parts or master gauges to verify clearances and interference fits meet drawing specifications, documenting any assembly sequence issues for future reference.

Pressure testing for pressure-bearing components involves hydrostatic testing at 1.5 times working pressure minimum, holding pressure for specified durations typically lasting 10 minutes. Visual inspection during and after pressure testing identifies leaks or deformation that could indicate structural issues. Load testing for structural components applies specified proof loads when required, measuring permanent deformation and verifying components return to original dimensions after load removal.

Decision Gate E determines whether components meet all functional requirements. Components passing functional tests proceed to final documentation, while those failing require root cause determination to evaluate whether issues stem from design problems or manufacturing defects.

Final Documentation and Disposition

Final documentation and disposition requires 15-20 minutes and must be completed by Quality Inspectors with final sign-off authority. Inspection package compilation includes complete inspection reports with all actual measurements, photographs of notable features or concerns, and attached material certifications with test results providing comprehensive documentation.

Certificate of Conformance generation summarizes all inspection results, confirms compliance with customer requirements, and obtains required signatures and quality stamps for traceability. Component disposition follows three categories: ACCEPT components receive green tags and move to shipping preparation, ACCEPT WITH DEVIATION

components receive yellow tags requiring customer notification, while REJECT components receive red tags for quarantine and disposition per non-conformance report processes.

Outputs Generated

The final inspection process generates comprehensive primary outputs including completed Inspection Reports using Form QC-TI-001A with all measurements and results, Certificates of Conformance signed by authorized quality representatives, accepted components properly tagged and ready for shipment, and updated Route Cards with final inspection completion stamps.

Secondary outputs include Non-Conformance Reports for rejected components, Process Improvement Suggestions based on recurring issues identified during inspection, Equipment Calibration Usage Log entries for all measuring instruments used, and Training Needs Assessments when inspector knowledge gaps are identified.

Personnel Responsibilities and Qualifications

Quality Technician Level II personnel perform dimensional inspections using precision measuring equipment, complete surface finish verification and visual inspections, and maintain calibration status of assigned measuring equipment. Required certifications include Level II Quality Inspector certification with CMM Programming preferred, and minimum 3 years titanium component inspection experience.

Senior Quality Inspectors review and approve all inspection reports before final sign-off, investigate and resolve complex measurement discrepancies, and interface with engineering on drawing interpretation questions. Required certifications include Senior Quality Inspector and GD&T Advanced certifications, with minimum 5 years aerospace quality experience and titanium expertise required.

Special Cases and Exception Handling

Rush Orders triggered by customer requests for expedited delivery with less than 24-hour lead time require dedicated inspector assignment with no other concurrent tasks, supervisor approval before starting inspection, customer notification of inspection completion within 2 hours, and premium rush inspection fees per company pricing structure.

Prototype Components representing first articles or design verification components require engineering representatives present during inspection, complete dimensional reports rather than sampling approaches, metallurgical cross-section analysis when requested by engineering, and design feedback documentation for future improvements.

Rework Components returning after customer rejection or internal rework require independent inspectors different from original inspectors performing re-inspection, 100% dimensional verification regardless of original sampling plans, root cause analysis documentation completion before re-inspection, and quality manager approval for final acceptance.

High-Value Components exceeding \$10,000 each require dual inspector verification with two independent inspections, quality engineering review of inspection plans before starting,

customer witness inspection options, and enhanced packaging and handling procedures post-inspection.

Critical Safety Components identified as flight-critical or life-sustaining medical devices require 100% inspection with no sampling allowed, traceability documentation to raw material heat lots, special handling procedures with dedicated storage areas, and quality manager signatures on Certificates of Conformance.

Environmental Condition Variations when temperature falls outside 18-22°C range or humidity exceeds 65% require documentation of actual environmental conditions on inspection reports, extended stabilization time with minimum 4 hours for components, re-calibration verification of precision measuring equipment, and engineering approval to proceed with inspection.

Standards and Training Requirements

Our operations reference comprehensive industry standards including AS9100D for Quality Management Systems Requirements for Aviation, Space and Defense Organizations, ISO 13485 for Medical Devices Quality Management Systems, ASTM B265 Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate, AMS 4911 for Titanium Alloy Sheet, Strip, and Plate 6Al-4V Annealed, Company Quality Manual QM-2024-Rev3, and Calibration Procedure CAL-001 for Dimensional Measuring Equipment.

All personnel performing titanium final inspection must complete comprehensive training including Company Quality Systems Training requiring 40 hours initially with 8 hours annual refresher training, Titanium Materials and Properties Course requiring 16 hours renewed every 3 years, GD&T Fundamentals requiring 24 hours initially with 8 hours every 2 years, CMM Operation and Programming requiring 32 hours with equipment-specific training, and Customer-Specific Requirements Training varying by customer with separate documentation.

Document Control and Revision History

This document was prepared by Sarah Martinez, Senior Quality Engineer, reviewed by Dr. James Chen, Materials Engineering Manager, and approved by Michael Rodriguez, Quality Assurance Director. Distribution includes all Quality Department Personnel, Production Supervisors, Engineering Department for reference, and Customer Quality Representatives upon request.

The revision history demonstrates continuous improvement with Rev 3.2 adding rush order procedures, Rev 3.1 clarifying CMM requirements for complex geometry, and Rev 3.0 representing a major revision incorporating AS9100D requirements.